

KAVYA SHIVAKUMAR

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PROFESSIONAL EXPERIENCE

Quality Assurance Intern - Carfax, USA

May 2025

- Improved test coverage by reviewing requirement specifications and technical design documents, ensuring alignment with functional expectations through collaborative discussions with developers, analysts, and product managers.
- Enhanced product quality by developing and executing detailed test plans and test cases for both web and mobile platforms.
- Increased test efficiency by identifying repetitive manual test cases and contributing to the automation and streamlining the QA workflow.
- Ensured product stability across releases by performing regression testing during various software development stages, validating both new and existing functionalities.
- Accelerated issue resolution by tracking, documenting, and reporting bugs during testing cycles, collaborating with developers to troubleshoot and validate fixes.
- Improved communication between teams by actively participating in sprint planning, QA reviews, and requirement analysis meetings to provide valuable testing feedback early in the development cycle.

Software Developer - Oracle India Pvt Ltd, Bangalore, India

July 2018 – August 2024

Worked on Application Development Framework:

- Refined product quality by 20% through streamlined CI/CD processes and the integration of automated testing tools.
- Performed Regression Testing at various phases by automating the scripts to reduce the time by 60% per script.

Engaged in Automation:

- Decreased manual testing efforts by 48% by designing and developing an automation framework integrated with Jenkins for regular automated test execution.
- Elevated test clarity and collaboration by 50% through BDD implementation using Cucumber and structured step definitions.
- Boosted automation performance by 40% by migrating existing Ant-based automation to a Maven project for improved build efficiency and dependency management.

Focused on Docker and Oracle Cloud Infrastructure:

- Enhanced team proficiency by leading internal training sessions after mastering and implementing Docker, Kubernetes, and JMeter for efficient containerization, orchestration, and performance testing.
- Reduced execution time and dependencies by 55% by building Docker images for Oracle ADF and deploying containers to run existing scripts and automation efficiently.
- Migrated scripts and automation from on-premises to Oracle Cloud infrastructure, reducing hardware usage by 61%.

EDUCATION

George Mason University, Fairfax, USA (GPA: 3.78/4)
MS in Computer Science

Expected Graduation Date: May 2026

RV College of Engineering, Bangalore, India (GPA: 8.65/10.0)
Bachelor of Engineering in Computer Science

August 2015 – July 2018

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Java, Python, Perl
- **Others:** Oracle Cloud Infrastructure, Docker, Kubernetes, AWS, MySQL, Oracle database, Jenkins, Selenium, JMeter, Git, Cucumber, JUnit, TestNG, REST API Testing, Microservices, JSON, XML, JavaScript, HTML, CSS, web services

PROJECTS

Supply Chain Management System | November 2024 - December 2024 – Python, cx_Oracle

GitHub [Link](#)

- Enabled seamless supply chain data operations by building a full-stack Flask web app with CRUD functionality, integrating cx_Oracle for real-time interaction with Oracle database tables like Items and Orders.
- Engineered an efficient user interface with HTML, CSS, and JavaScript enabling real-time data handling through GET, POST, and DELETE methods.

Brain Tumor Segmentation Using Convolutional Neural Networks in MRI images | December 2017 - May 2018 - Python, Convolutional Neural Network.

- Minimized manual clinical time by 60–70% by engineering a brain tumor segmentation model using CNNs with layered spatial pattern extraction.
- Delivered research project ahead of schedule with 7.2% higher accuracy by leading planning and optimizing deep learning models for MRI image analysis.

Theft Monitoring System | August 2017 – November 2017 – Java, Android Studio.

- Accelerated device recovery by 60% by formulating an app to send GPS-based last known location via SMS to an alternate number.
- Achieved approximately 70% efficiency by integrating Android's Location Services, SMS Manager, and Broadcast Receivers to monitor SIM changes.

Sonar Image Processing | February 2016 – May 2016 – Python

- Automated object detection in sonar images by developing a system requiring no human intervention for analysis.
- Enhanced detection accuracy to approximately 85% by applying median filtering, Gaussian smoothing, and adaptive filtering to clean noisy sonar data.

CERTIFICATION

- Oracle Cloud Infrastructure (OCI) 2025 Foundations Associate certification.
- Oracle Cloud Infrastructure (OCI) 2024 Foundations Associate certification.